Theoretical Background

Geometry

The physical domain consists of two infinite parallel plates separated by a constant distance H. The coordinate system is defined such that the x-axis is aligned with the streamwise flow direction, and the y-axis is normal to the plates.

- **Domain:** $0 \leq y \leq H$ and $-\infty < x < \infty$
- **Top Plate ($y = H$):** Stationary (velocity $U = 0$)
- **Bottom Plate ($y = 0$):** Moving with a constant velocity U_w

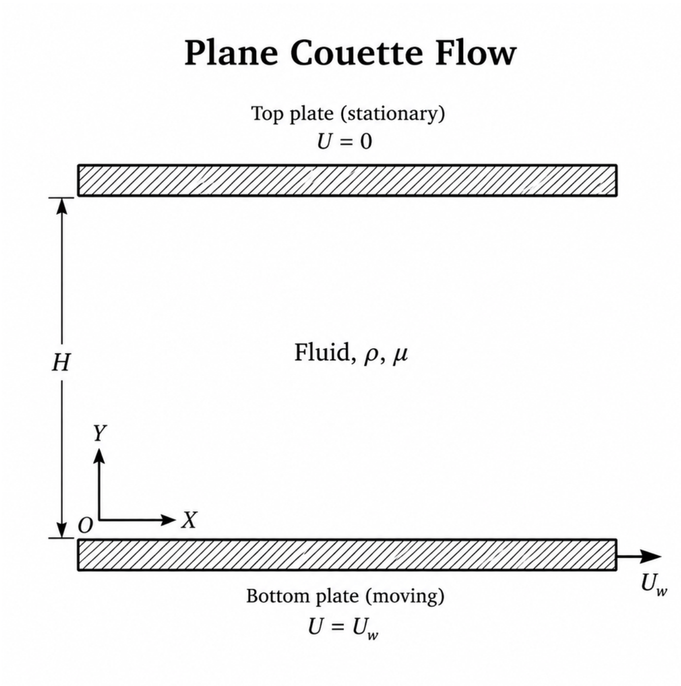

Figure 1: Channel Geometry

As illustrated in Figure 1, the fluid occupies the region between the two parallel plates, with the origin of the coordinate system located on the bottom plate.

Governing Equations

To derive the mathematical model for plane Couette flow, the following assumptions are adopted:

1. The fluid is incompressible, and its density ρ is constant.
2. The thermophysical properties of the fluid are constant, including the dynamic viscosity μ.
3. The flow is one-dimensional; therefore, the velocity has only an x-component.
4. The plates are infinitely long in the flow direction.

5. No external pressure gradient is imposed in the streamwise direction, such that $\partial p/\partial x = 0$.
6. Gravitational effects and other body forces are neglected.

Continuity Equation

The general form of the continuity equation is:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0 \tag{1}$$

For a two-dimensional Cartesian coordinate system, Equation (1) is expressed as:

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} = 0 \tag{2}$$

Given that the density is constant, $\partial \rho/\partial t = 0$, Equation (2) reduces to:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{3}$$

For the one-dimensional plane Couette flow considered here, the velocity field has only a streamwise component:

$$\mathbf{V} = [u(y,t),\, 0]$$

Therefore, the velocity component normal to the plates is identically zero:

$$v = 0$$

Thus, Equation (3) simplifies to:

$$\frac{\partial u}{\partial x} = 0 \tag{4}$$

It follows from Equation (4) that the streamwise velocity is independent of x and can be written as:

$$u = u(y,t)$$

Momentum Equation

For an incompressible Newtonian fluid with body-force effects neglected, the Navier–Stokes momentum equation is:

$$\rho\left(\frac{\partial \mathbf{V}}{\partial t}+\mathbf{V}\cdot\nabla\mathbf{V}\right)=-\nabla p+\mu\nabla^2\mathbf{V} \tag{5}$$

Given the velocity field $\mathbf{V}=[u(y,t),\,0]$, the x-momentum component becomes:

$$\rho\left(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}\right)=-\frac{\partial p}{\partial x}+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right) \tag{6}$$

Substituting the kinematic conditions:

$$\frac{\partial u}{\partial x}=0, \qquad v=0, \qquad \frac{\partial^2 u}{\partial x^2}=0$$

together with the zero streamwise pressure gradient:

$$\frac{\partial p}{\partial x}=0$$

into Equation (6) yields:

$$\rho\frac{\partial u}{\partial t}=\mu\frac{\partial^2 u}{\partial y^2} \tag{7}$$

Dividing Equation (7) by the density gives the one-dimensional transient diffusion equation:

$$\frac{\partial u}{\partial t}=\nu\frac{\partial^2 u}{\partial y^2} \tag{8}$$

where:

$$\nu=\frac{\mu}{\rho}$$

is the kinematic viscosity.

Additionally, the y-momentum equation reduces to:

$$\frac{\partial p}{\partial y}=0$$

which confirms that the pressure is uniform across the channel height.

Analytical Solution

(This section is reserved for the analytical solution comparison.)

Initial and Boundary Conditions

The initial and boundary conditions for the transient plane Couette flow problem are defined as follows.

Initial Condition

At the initial time $t = 0$, the fluid is at rest throughout the channel interior, while the bottom plate is already moving with the constant velocity U_w:

$$u(0,0) = U_w$$

$$u(y,0) = 0 \qquad \text{for } 0 < y \leq H$$

Boundary Conditions

For $t \geq 0$, the velocity field satisfies the following Dirichlet boundary conditions:

$$u(0,t) = U_w$$

$$u(H,t) = 0$$

The first condition represents the no-slip condition at the moving bottom plate, while the second condition represents the no-slip condition at the stationary top plate. Together, these conditions describe the impulsive start of the bottom plate and the subsequent transient diffusion of momentum across the channel.

Domain Discretization

To solve the governing diffusion equation numerically, the continuous spatial domain $0 \leq y \leq H$ is discretized into a finite number of grid points (nodes). Let N represent the total number of nodes along the y-direction. The uniform distance between consecutive nodes, referred to as the grid spacing Δy, is defined as:

$$\Delta y = \frac{H}{N-1}$$

The discrete spatial coordinates of these grid points are given by:

$$y_j = j\Delta y, \quad j = 0, 1, 2, \dots, N-1$$

In alignment with C++ indexing standards, the index j starts from 0 (bottom wall) and ends at $N-1$ (top wall). The spatial discretization of the domain is illustrated in Figure 2.

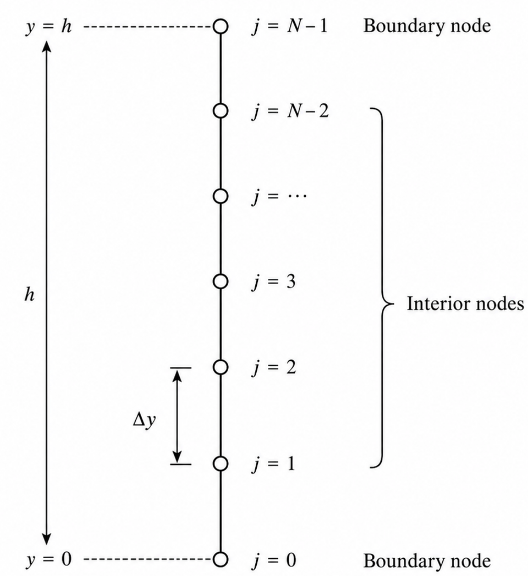

Figure 2: Computational Grid

Similarly, the temporal domain is discretized into uniform intervals of Δt, where n denotes the time level. The continuous velocity $u(y,t)$ at a specific grid location y_j and time $t^n = n\Delta t$ is represented using the following discrete notation:

$$u(y_j, t^n) \approx u_j^n$$

This discrete representation serves as the foundation for approximating the partial derivatives in the governing equation using Finite Difference Methods (FDM).

Explicit DuFort-Frankel Scheme

The DuFort-Frankel scheme is a modified form of the Richardson method developed to improve the stability characteristics of explicit time-marching schemes for the one-dimensional diffusion equation.

The DuFort-Frankel method modifies the spatial discretization by replacing the central value at the current time level, u_j^n, with the average of the same spatial node at the adjacent time levels:

$$u_j^n \approx \frac{u_j^{n+1} + u_j^{n-1}}{2}$$

This modification introduces the unknown value u_j^{n+1} into the spatial approximation. However, after rearrangement, the resulting equation remains explicit because u_j^{n+1} can be calculated directly from previously known neighboring values.

The governing one-dimensional diffusion equation is evaluated at the grid point (y_j, t^n), giving:

$$\left(\frac{\partial u}{\partial t}\right)_j^n = \nu \left(\frac{\partial^2 u}{\partial y^2}\right)_j^n \tag{9}$$

The DuFort-Frankel scheme uses:

- a central difference in time;
- a modified central difference in space;
- three consecutive time levels: $n-1$, n, and $n+1$;
- a direct explicit update after algebraic rearrangement.

A. Temporal Discretization of the DuFort-Frankel Scheme

The temporal derivative is approximated using the same central difference employed in the Richardson scheme. The first-order temporal derivative at (y_j, t^n) is:

$$\left(\frac{\partial u}{\partial t}\right)_j^n = \frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} + \mathcal{O}\left((\Delta t)^2\right) \tag{10}$$

Thus, the central temporal approximation is formally second-order accurate in time.

The same temporal approximation is also used by the Richardson scheme. Therefore, the principal difference between the Richardson and DuFort-Frankel methods is not the temporal discretization, but the modified treatment of the central spatial value.

B. Modified Spatial Discretization

The conventional second-order central approximation for the second-order spatial derivative is:

$$\left(\frac{\partial^2 u}{\partial y^2}\right)_j^n = \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta y)^2} + \mathcal{O}\left((\Delta y)^2\right) \tag{11}$$

This is the same spatial approximation used by the FTCS and Richardson schemes.

However, directly using u_j^n in Equation (11) leads to the Richardson formulation. The DuFort-Frankel method replaces the central value u_j^n with the average of the values at the same spatial node at time levels $n+1$ and $n-1$:

$$u_j^n \approx \frac{u_j^{n+1} + u_j^{n-1}}{2} \tag{12}$$

Substituting Equation (12) into the conventional central approximation in Equation (11) gives the modified spatial approximation:

$$\left(\frac{\partial^2 u}{\partial y^2}\right)_j^n \approx \frac{u_{j+1}^n - \left(u_j^{n+1} + u_j^{n-1}\right) + u_{j-1}^n}{(\Delta y)^2} + \mathcal{O}\left((\Delta y)^2\right) \tag{13}$$

The key modification relative to the Richardson method is therefore:

$$-2u_j^n \quad \longrightarrow \quad -\left(u_j^{n+1} + u_j^{n-1}\right)$$

In this way, the current-time central value is replaced by values from the two adjacent time levels.

C. Discrete Algebraic Equation

Substituting the central temporal approximation from Equation (10) and the modified spatial approximation from Equation (13) into the governing diffusion equation in Equation (9) gives:

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} = \nu \left(\frac{u_{j+1}^n - u_j^{n+1} - u_j^{n-1} + u_{j-1}^n}{(\Delta y)^2}\right) \tag{14}$$

Defining the dimensionless diffusion number as:

$$d = \frac{\nu \Delta t}{(\Delta y)^2} \tag{15}$$

and multiplying Equation (14) by $2\Delta t$, we obtain:

$$u_j^{n+1} - u_j^{n-1} = 2d\left(u_{j+1}^n - u_j^{n+1} - u_j^{n-1} + u_{j-1}^n\right) \tag{16}$$

Collecting the terms at time levels $n+1$ and $n-1$ gives:

$$(1+2d)u_j^{n+1} = (1-2d)u_j^{n-1} + 2d\left(u_{j+1}^n + u_{j-1}^n\right) \tag{17}$$

Finally, dividing Equation (17) by $(1{+}2d)$ yields the explicit DuFort-Frankel update equation:

$$u_j^{n+1} = \frac{(1-2d)u_j^{n-1} + 2d\left(u_{j+1}^n + u_{j-1}^n\right)}{1+2d} \tag{18}$$

Equation (18) is the final algebraic form of the DuFort-Frankel scheme.

Unlike the Richardson equation, the central value u_j^n does not appear explicitly in the update equation. Instead, the current-time central value has been replaced by the average defined in Equation (12).

For a grid containing N nodes, Equation (18) is applied at all interior nodes:

$$j = 1, 2, \dots, N-2$$

The boundary values at $j = 0$ and $j = N-1$ are assigned directly from the prescribed physical boundary conditions.

Because Equation (18) directly provides u_j^{n+1} from known values at time levels n and $n-1$, the DuFort-Frankel method is classified as an explicit three-time-level scheme. Therefore, no tridiagonal or other algebraic system must be solved at each time step.

D. Comparison with the FTCS and Richardson Schemes

The three explicit schemes considered so far differ in their temporal discretization, spatial treatment, and stability properties.

The FTCS scheme uses a forward difference in time and a conventional central difference in space. Its update equation is:

$$u_j^{n+1} = u_j^n + d\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right) \tag{19}$$

The Richardson scheme uses central differences in both time and space. Its update equation is:

$$u_j^{n+1} = u_j^{n-1} + 2d\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right)$$

The DuFort-Frankel scheme retains the central temporal difference of Richardson but replaces the current-time central spatial value with the average in Equation (12), resulting in Equation (18).

The principal differences among the three explicit schemes are summarized below:

Scheme	Temporal discretization	Spatial discretization	Time levels	Formal accuracy	Stability
FTCS	Forward, first-order	Central, second-order	2	First-order in time and second-order in space	Conditionally stable, $d \leq \frac{1}{2}$
Richardson	Central, second-order	Central, second-order	3	Second-order in time and space	Unconditionally unstable
DuFort-Frankel	Central, second-order	Modified central	3	Nominally second-order spatial treatment	Unconditionally stable for the diffusion equation

The comparison shows that the Richardson scheme and the DuFort-Frankel scheme share the same central temporal discretization and three-time-level structure. Their main difference is the treatment of the central spatial value:

- Richardson uses u_j^n directly.
- DuFort-Frankel replaces u_j^n with $\frac{1}{2}(u_j^{n+1} + u_j^{n-1})$.

This apparently small modification changes the stability behavior substantially. The Richardson method is unconditionally unstable, whereas the DuFort-Frankel method is unconditionally stable for the one-dimensional diffusion equation.

Compared with FTCS, DuFort-Frankel removes the restrictive condition $d \leq \frac{1}{2}$. Consequently, larger time steps can be used without the classical FTCS instability. However, unconditional stability does not imply unlimited accuracy. Very large values of d may still introduce excessive numerical damping or temporal distortion.

E. Stability Analysis of the DuFort-Frankel Scheme

The stability of the DuFort-Frankel scheme can be examined using a von Neumann Fourier analysis. Consider a Fourier error mode of the form:

$$\epsilon_j^n = G^n e^{ij\theta} \tag{20}$$

where G is the amplification factor, θ is the phase angle, and $i = \sqrt{-1}$.

Substituting Equation (20) into the update equation in Equation (18) gives:

$$(1+2d)G^{n+1}e^{ij\theta} = (1-2d)G^{n-1}e^{ij\theta} + 2dG^n \left(e^{i(j+1)\theta} + e^{i(j-1)\theta}\right) \tag{21}$$

Dividing Equation (21) by $G^{n-1}e^{ij\theta}$ gives:

$$(1+2d)G^2 = (1-2d) + 2dG\left(e^{i\theta} + e^{-i\theta}\right) \tag{22}$$

Using the identity $e^{i\theta} + e^{-i\theta} = 2\cos\theta$, Equation (22) becomes:

$$(1+2d)G^2 - 4d\cos\theta\, G - (1-2d) = 0 \tag{23}$$

The two amplification-factor roots are:

$$G_{\pm} = \frac{2d\cos\theta \pm \sqrt{1-4d^2\sin^2\theta}}{1+2d} \tag{24}$$

When $1-4d^2\sin^2\theta \geq 0$, the roots are real. When $1-4d^2\sin^2\theta < 0$, the roots form a complex-conjugate pair. In the latter case, the squared magnitude of each root is:

$$\left|G_{\pm}\right|^2 = \frac{2d-1}{2d+1} < 1, \qquad d > \frac{1}{2} \tag{25}$$

For the real-root case, the roots also satisfy:

$$\left|G_{\pm}\right| \leq 1 \tag{26}$$

Therefore, all Fourier modes remain bounded for every positive diffusion number:

$$\left|G_{\pm}\right| \leq 1, \qquad d > 0 \tag{27}$$

Thus, the DuFort-Frankel scheme is unconditionally stable for the one-dimensional diffusion equation. This is the principal advantage of DuFort-Frankel over both FTCS and Richardson:

- FTCS is stable only when $d \leq \frac{1}{2}$.
- Richardson is unstable for every $d > 0$.
- DuFort-Frankel remains stable for every positive d.

Nevertheless, unconditional stability does not mean that an arbitrarily large time step produces an accurate solution. The value of Δt should still be selected according to the desired temporal resolution and accuracy.

F. Initialization of the Three-Time-Level Scheme

The DuFort-Frankel method uses the three time levels $n-1$, n, and $n+1$. Therefore, the solution must be available at two initial time levels before Equation (18) can be applied.

The initial condition provides the solution at time level $n = 0$. The solution at time level $n = 1$ is then required to start the three-time-level DuFort-Frankel update.

A common initialization approach is to compute u_j^1 using the FTCS scheme:

$$u_j^1 = u_j^0 + d\left(u_{j+1}^0 - 2u_j^0 + u_{j-1}^0\right) \tag{28}$$

However, this FTCS start-up step must itself satisfy the FTCS stability condition. Therefore, the corresponding diffusion number must obey:

$$d \leq \frac{1}{2} \tag{29}$$

If the desired global time step is too large to satisfy Equation (29), the first time interval should be divided into smaller substeps. In that case, u_j^1 is obtained by applying FTCS repeatedly with a reduced sub-time-step Δt_{sub} such that:

$$d_{\text{sub}} = \frac{\nu \Delta t_{\text{sub}}}{(\Delta y)^2} \leq \frac{1}{2} \tag{30}$$

This substepping procedure ensures that the initialization remains numerically stable. It is important because an unstable or overly diffusive FTCS start-up can contaminate the initial value u_j^1, and that error is then propagated into the subsequent DuFort-Frankel time levels.

In other words, although the DuFort-Frankel scheme itself is unconditionally stable for the diffusion equation, its first computed time level still acts as input data for all later steps. Therefore, a poor initialization may degrade the accuracy of the entire solution, even if the DuFort-Frankel marching formula remains stable.

For this reason, using FTCS only as a start-up method is acceptable only when its stability restriction can be respected. Otherwise, it is preferable to generate u_j^1 using one of the implicit methods introduced in the following sections, such as the Implicit Euler (Laasonen) scheme or the Crank-Nicolson scheme. This avoids the FTCS start-up restriction entirely and provides a more robust initialization for the DuFort-Frankel method.

After u_j^1 has been obtained, the DuFort-Frankel update in Equation (18) is applied for subsequent time levels.

G. Final Comparison of the Explicit Schemes

The final comparison of the three explicit methods is summarized in the following table:

Property	FTCS	Richardson	DuFort-Frankel
Temporal discretization	Forward difference	Central difference	Central difference
Spatial discretization	Conventional central difference	Conventional central difference	Modified central difference
Number of time levels	Two	Three	Three

Property	FTCS	Richardson	DuFort-Frankel
Explicit formulation	Yes	Yes	Yes
Temporal accuracy	First-order	Second-order	Central, formally second-order
Spatial accuracy	Second-order	Second-order	Central spatial treatment
Stability condition	$d \leq \frac{1}{2}$	No stable positive d	Unconditionally stable
Need for initialization	No additional time level	Yes	Yes
Practical use in this project	Suitable with time-step restriction	Not suitable	Suitable
Main limitation	Restricted time step	Unconditional instability	Possible numerical damping for large d

The three methods illustrate an important principle in numerical analysis: formal accuracy and stability must be considered together.

The FTCS scheme is simple and accurate in space, but its time step is restricted by:

$$d \leq \frac{1}{2}$$

The Richardson scheme improves the formal temporal order from first to second order, but this improvement is accompanied by unconditional instability. Therefore, Richardson is not suitable for practical computation despite its high formal accuracy.

The DuFort-Frankel scheme modifies the Richardson formulation by replacing the current-time central value with an average of the adjacent time levels. This modification preserves an explicit update structure while eliminating the unconditional instability of Richardson.

Consequently, among the three explicit schemes considered in this report:

- FTCS is appropriate when the stability restriction is acceptable.
- Richardson is included for comparison but is not used in the numerical implementation.
- DuFort-Frankel is attractive when larger time steps are desired because it is unconditionally stable for the one-dimensional diffusion equation.

For the present transient Plane Couette Flow problem, the FTCS and DuFort-Frankel schemes provide practical explicit alternatives, whereas the Richardson scheme serves primarily as a theoretical comparison demonstrating that higher formal accuracy does not necessarily guarantee numerical stability.

Linear_Solvers

Numerical Schemes

In this section, the transition from the continuous partial differential equation to discrete algebraic forms suitable for C++ implementation is presented.

To solve the one-dimensional transient diffusion equation, previously introduced in Theoretical Background as Equation (31):

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2} \tag{31}$$

the continuous temporal and spatial derivatives must be approximated using finite difference expressions. Implicit numerical schemes are therefore introduced to obtain discrete formulations of the governing equation.

Implicit Numerical Schemes

1. **Laasonen Scheme**
2. **CrankNicolson Scheme**

Explicit FTCS Scheme

To solve the one-dimensional diffusion equation numerically, the continuous partial derivatives must be approximated using discrete finite difference expressions. The Forward-Time Central-Space (FTCS) scheme is the fundamental explicit method used for this purpose.

The governing diffusion equation is evaluated at the grid point (y_j, t^n). Using the spatial index j and the temporal index n, one-dimensional diffusion equation can be written in discrete-point notation as:

$$\left(\frac{\partial u}{\partial t}\right)_j^n = \nu \left(\frac{\partial^2 u}{\partial y^2}\right)_j^n \tag{32}$$

The FTCS scheme employs a first-order forward difference in time and a second-order central difference in space.

A. Discretization of the First-Order Temporal Derivative

To approximate the first-order temporal derivative $\left(\frac{\partial u}{\partial t}\right)_j^n$, a Taylor series expansion of $u(y_j, t^{n+1})$ about the point (y_j, t^n) is used:

$$u_j^{n+1} = u_j^n + \left(\frac{\partial u}{\partial t}\right)_j^n \Delta t + \frac{1}{2!}\left(\frac{\partial^2 u}{\partial t^2}\right)_j^n (\Delta t)^2 + \frac{1}{3!}\left(\frac{\partial^3 u}{\partial t^3}\right)_j^n (\Delta t)^3 + \cdots \tag{33}$$

Rearranging Equation (33) to isolate the first-order temporal derivative gives:

$$\left(\frac{\partial u}{\partial t}\right)_j^n = \frac{u_j^{n+1} - u_j^n}{\Delta t} + \mathcal{O}(\Delta t) \tag{34}$$

The term $\mathcal{O}(\Delta t)$ represents the truncation error of the temporal approximation. Therefore, the forward difference approximation is first-order accurate in time. If the time step Δt is reduced by half, the temporal truncation error is also approximately reduced by half.

B. Discretization of the Second-Order Spatial Derivative

To approximate the second-order spatial derivative $\left(\frac{\partial^2 u}{\partial y^2}\right)_j^n$, the values u_{j+1}^n and u_{j-1}^n are expanded about the point (y_j, t^n).

Forward Spatial Taylor Expansion

The Taylor series expansion of u_{j+1}^n about the point (y_j, t^n) is:

$$u_{j+1}^n = u_j^n + \left(\frac{\partial u}{\partial y}\right)_j^n \Delta y + \frac{1}{2!}\left(\frac{\partial^2 u}{\partial y^2}\right)_j^n (\Delta y)^2 + \frac{1}{3!}\left(\frac{\partial^3 u}{\partial y^3}\right)_j^n (\Delta y)^3 + \frac{1}{4!}\left(\frac{\partial^4 u}{\partial y^4}\right)_j^n (\Delta y)^4 + \cdots \tag{35}$$

Backward Spatial Taylor Expansion

Similarly, the Taylor series expansion of u_{j-1}^n about the point (y_j, t^n) is:

$$u_{j-1}^n = u_j^n - \left(\frac{\partial u}{\partial y}\right)_j^n \Delta y + \frac{1}{2!}\left(\frac{\partial^2 u}{\partial y^2}\right)_j^n (\Delta y)^2 - \frac{1}{3!}\left(\frac{\partial^3 u}{\partial y^3}\right)_j^n (\Delta y)^3 + \frac{1}{4!}\left(\frac{\partial^4 u}{\partial y^4}\right)_j^n (\Delta y)^4 - \cdots \tag{36}$$

Adding Equations (35) and (36) eliminates the odd-order spatial derivative terms:

$$u_{j+1}^n + u_{j-1}^n = 2u_j^n + \left(\frac{\partial^2 u}{\partial y^2}\right)_j^n (\Delta y)^2 + \frac{1}{12}\left(\frac{\partial^4 u}{\partial y^4}\right)_j^n (\Delta y)^4 + \cdots \tag{37}$$

The fourth-order and higher-order terms can be included in the truncation error. Therefore, rearranging Equation (37) to isolate the second-order spatial derivative gives:

$$\left(\frac{\partial^2 u}{\partial y^2}\right)_j^n = \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta y)^2} + \mathcal{O}\left((\Delta y)^2\right) \tag{38}$$

This central difference approximation is second-order accurate in space. Consequently, reducing the grid spacing Δy by half reduces the spatial truncation error by approximately a factor of four.

C. Discrete Algebraic Equation

Substituting the first-order forward difference approximation from Equation (34) and the second-order central difference approximation from Equation (38) into the governing diffusion equation in Equation (32) yields:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \nu \left(\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta y)^2}\right) \tag{39}$$

Multiplying both sides of Equation (39) by Δt gives:

$$u_j^{n+1} - u_j^n = \frac{\nu \Delta t}{(\Delta y)^2}\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right) \tag{40}$$

Rearranging Equation (40) to isolate the unknown value at the new time level, u_j^{n+1}, results in:

$$u_j^{n+1} = u_j^n + \frac{\nu \Delta t}{(\Delta y)^2}\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right) \tag{41}$$

The dimensionless diffusion number is defined as:

$$d = \frac{\nu \Delta t}{(\Delta y)^2} \tag{42}$$

In heat-transfer applications, this parameter is commonly referred to as the Fourier number. Using the diffusion number defined in Equation (42), the FTCS update equation becomes:

$$u_j^{n+1} = u_j^n + d\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right) \tag{43}$$

Expanding the terms in Equation (43), the explicit FTCS formulation can also be written as:

$$u_j^{n+1} = d\, u_{j-1}^n + (1 - 2d)\, u_j^n + d\, u_{j+1}^n \tag{44}$$

This form clearly shows the contribution of the three neighboring nodes at the previous time level.

The computational molecule, or stencil, for the FTCS scheme is illustrated in Figure 3. The unknown value at the new time level $n+1$ is explicitly calculated using three known values from the previous time level n.

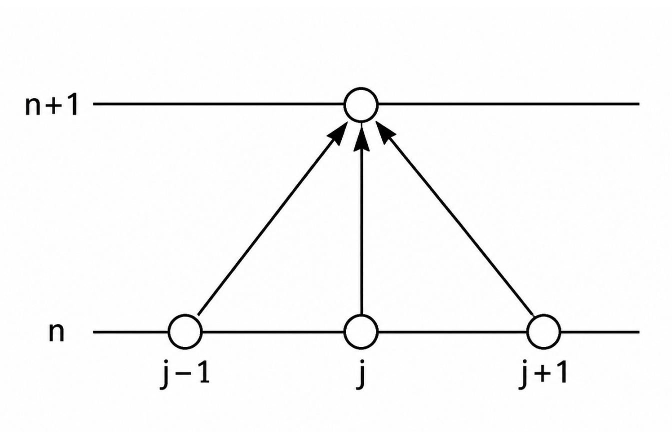

Figure 3: Computational stencil for the explicit FTCS scheme

Equation (44) is an explicit algebraic formulation because the value at the new time level depends only on values that are already known at the previous time level. Therefore, no system of algebraic equations must be solved at each time step.

For a grid containing N nodes, the FTCS update equation is applied at all interior nodes:

$$j = 1, 2, \dots, N-2$$

The boundary-node values at $j = 0$ and $j = N-1$ are assigned directly from the prescribed physical boundary conditions.

$$\mathcal{T}_{\text{FTCS}} = \mathcal{O}(\Delta t) + \mathcal{O}\left((\Delta y)^2\right) \tag{45}$$

D. Accuracy of the FTCS Scheme

The temporal and spatial truncation errors obtained from Equations (34) and (38) are:

$$\text{Temporal truncation error} = \mathcal{O}(\Delta t)$$

$$\text{Spatial truncation error} = \mathcal{O}\left((\Delta y)^2\right)$$

Therefore, the FTCS scheme is:

- first-order accurate in time;
- second-order accurate in space.

The combined local truncation error can be expressed as:

E. Stability Analysis of the Explicit FTCS Scheme

The explicit FTCS scheme is conditionally stable. This means that the numerical solution remains stable only if the time step Δt, grid spacing Δy, and diffusion coefficient ν satisfy a specific stability condition.

For the one-dimensional diffusion equation, the von Neumann stability analysis gives the following restriction on the diffusion number:

$$d = \frac{\nu \Delta t}{(\Delta y)^2} \leq \frac{1}{2} \tag{46}$$

Equivalently, the stability condition can be written as:

$$0 \leq d \leq \frac{1}{2} \tag{47}$$

When $d > \frac{1}{2}$, high-frequency numerical modes are amplified rather than dissipated. As a result, round-off and truncation errors may grow with successive time steps, eventually producing an unstable and non-physical solution.

For a fixed grid spacing Δy, the maximum permissible time step is obtained from the stability condition in Equation (46):

$$\Delta t_{\max} = \frac{(\Delta y)^2}{2\nu} \tag{48}$$

In practical implementations, a safety factor is often used. For example, selecting $d = 0.45$ instead of the limiting value $d = 0.5$ provides a small numerical safety margin:

$$\Delta t = d\frac{(\Delta y)^2}{\nu}, \qquad d < \frac{1}{2} \tag{49}$$

Thus, the time step used in the C++ implementation must be selected so that the diffusion number satisfies the FTCS stability restriction in Equation (46). If the grid is refined while ν remains constant, the allowable time step decreases proportionally to $(\Delta y)^2$.

Explicit Richardson Scheme

As a second explicit approach, the Richardson scheme can be applied to the one-dimensional transient diffusion equation. Unlike the FTCS scheme, which uses a forward difference in time, the Richardson scheme employs a central difference in time and a central difference in space.

The central temporal discretization provides second-order accuracy in time. However, despite its second-order accuracy in both time and space, the Richardson scheme is unconditionally unstable for the one-dimensional diffusion equation. Therefore, it has no practical value for the present transient Plane Couette Flow problem.

The governing diffusion equation is evaluated at the grid point (y_j, t^n), giving:

$$\left(\frac{\partial u}{\partial t}\right)_j^n = \nu \left(\frac{\partial^2 u}{\partial y^2}\right)_j^n \tag{50}$$

The Richardson scheme uses:

- a second-order central difference in time;
- a second-order central difference in space;
- three consecutive time levels: $n-1$, n, and $n+1$.

A. Discretization of the First-Order Temporal Derivative

To approximate the first-order temporal derivative $\left(\frac{\partial u}{\partial t}\right)_j^n$, Taylor series expansions of u_j^{n+1} and u_j^{n-1} are developed about the point (y_j, t^n).

Forward Temporal Taylor Expansion

The Taylor series expansion of $u(y_j, t^{n+1})$ about (y_j, t^n) is:

$$u_j^{n+1} = u_j^n + \left(\frac{\partial u}{\partial t}\right)_j^n \Delta t + \frac{1}{2!}\left(\frac{\partial^2 u}{\partial t^2}\right)_j^n (\Delta t)^2 + \frac{1}{3!}\left(\frac{\partial^3 u}{\partial t^3}\right)_j^n (\Delta t)^3 + \frac{1}{4!}\left(\frac{\partial^4 u}{\partial t^4}\right)_j^n (\Delta t)^4 + \cdots \tag{51}$$

Backward Temporal Taylor Expansion

Similarly, the Taylor series expansion of $u(y_j, t^{n-1})$ about (y_j, t^n) is:

$$u_j^{n-1} = u_j^n - \left(\frac{\partial u}{\partial t}\right)_j^n \Delta t + \frac{1}{2!}\left(\frac{\partial^2 u}{\partial t^2}\right)_j^n (\Delta t)^2 - \frac{1}{3!}\left(\frac{\partial^3 u}{\partial t^3}\right)_j^n (\Delta t)^3 + \frac{1}{4!}\left(\frac{\partial^4 u}{\partial t^4}\right)_j^n (\Delta t)^4 - \cdots \tag{52}$$

Subtracting Equation (52) from Equation (51) eliminates the even-order temporal derivative terms:

$$u_j^{n+1} - u_j^{n-1} = 2\left(\frac{\partial u}{\partial t}\right)_j^n \Delta t + \frac{1}{3}\left(\frac{\partial^3 u}{\partial t^3}\right)_j^n (\Delta t)^3 + \mathcal{O}\left((\Delta t)^5\right) \tag{53}$$

Rearranging Equation (53) to isolate the first-order temporal derivative gives:

$$\left(\frac{\partial u}{\partial t}\right)_j^n = \frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} + \mathcal{O}\left((\Delta t)^2\right) \tag{54}$$

The central difference approximation is therefore second-order accurate in time. The temporal truncation error is $\mathcal{O}\left((\Delta t)^2\right)$.

B. Discretization of the Second-Order Spatial Derivative

For the second-order spatial derivative, the same central difference approximation used in the FTCS scheme is applied. The Taylor expansions about (y_j, t^n) lead to:

$$\left(\frac{\partial^2 u}{\partial y^2}\right)_j^n = \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta y)^2} + \mathcal{O}\left((\Delta y)^2\right) \tag{55}$$

The spatial approximation is second-order accurate because its truncation error is $\mathcal{O}\left((\Delta y)^2\right)$.

C. Discrete Algebraic Equation

Substituting the central temporal approximation from Equation (54) and the central spatial approximation from Equation (55) into the governing equation in Equation (50) gives:

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} = \nu\left(\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta y)^2}\right) \tag{56}$$

Multiplying both sides of Equation (56) by $2\Delta t$ yields:

$$u_j^{n+1} - u_j^{n-1} = \frac{2\nu\Delta t}{(\Delta y)^2}\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right) \tag{57}$$

Rearranging Equation (57) to isolate the unknown value u_j^{n+1} gives:

$$u_j^{n+1} = u_j^{n-1} + \frac{2\nu\Delta t}{(\Delta y)^2}\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right) \tag{58}$$

The dimensionless diffusion number is defined as:

$$d = \frac{\nu\Delta t}{(\Delta y)^2} \tag{59}$$

Using the diffusion number defined in Equation (59), the Richardson scheme becomes:

$$u_j^{n+1} = u_j^{n-1} + 2d\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right) \tag{60}$$

Equation (60) is the final explicit Richardson update equation. It calculates the unknown value at time level $n+1$ using the known solution values at time levels n and $n-1$.

For a grid containing N nodes, Equation (60) is applied at all interior nodes:

$$j = 1, 2, \dots, N-2$$

The boundary values at $j = 0$ and $j = N-1$ are assigned directly from the prescribed physical boundary conditions.

Because Equation (60) uses two previous time levels, the Richardson method is classified as an explicit three-time-level scheme. No algebraic system needs to be solved at each time step.

D. Accuracy of the Richardson Scheme

According to Equation (54), the temporal discretization is second-order accurate:

$$\text{Temporal truncation error} = \mathcal{O}\left((\Delta t)^2\right)$$

According to Equation (55), the spatial discretization is also second-order accurate:

$$\text{Spatial truncation error} = \mathcal{O}\left((\Delta y)^2\right)$$

Therefore, the Richardson scheme is second-order accurate in both time and space. Its combined local truncation error can be expressed as:

$$\mathcal{T}_{\text{Richardson}} = \mathcal{O}\left((\Delta t)^2\right) + \mathcal{O}\left((\Delta y)^2\right) \tag{61}$$

Although this formal accuracy is higher than the first-order temporal accuracy of the FTCS scheme, accuracy alone does not guarantee numerical stability.

E. Stability Analysis of the Explicit Richardson Scheme

Despite its second-order accuracy in both time and space, the Richardson scheme is unconditionally unstable for the one-dimensional diffusion equation.

This behavior can be demonstrated using a von Neumann Fourier stability analysis. Consider a Fourier error mode of the form:

$$\epsilon_j^n = G^n e^{ij\theta} \tag{62}$$

where G is the amplification factor, θ is the phase angle, and $i = \sqrt{-1}$.

Substituting the Fourier mode from Equation (62) into the Richardson update equation in Equation (60) gives:

$$G^{n+1} e^{ij\theta} = G^{n-1} e^{ij\theta} + 2dG^n \left(e^{i(j+1)\theta} - 2e^{ij\theta} + e^{i(j-1)\theta} \right) \tag{63}$$

Dividing Equation (63) by $G^{n-1} e^{ij\theta}$ gives:

$$G^2 = 1 + 2dG \left(e^{i\theta} - 2 + e^{-i\theta} \right) \tag{64}$$

Using the identities $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ and $\cos\theta - 1 = -2\sin^2\left(\frac{\theta}{2}\right)$, Equation (64) becomes:

$$G^2 + 8d \sin^2\left(\frac{\theta}{2}\right) G - 1 = 0 \tag{65}$$

The two roots of the characteristic equation in Equation (65) are:

$$G_{\pm} = -4d\sin^2\left(\frac{\theta}{2}\right) \pm \sqrt{1 + 16d^2 \sin^4\left(\frac{\theta}{2}\right)} \tag{66}$$

For every $d > 0$ and every nonzero wave number, one of these roots has a magnitude greater than unity. In particular, the negative root satisfies:

$$|G_{-}| = 4d\sin^2\left(\frac{\theta}{2}\right) + \sqrt{1 + 16d^2 \sin^4\left(\frac{\theta}{2}\right)} > 1 \tag{67}$$

Therefore, at least one numerical error mode grows with successive time levels for any positive diffusion number d. The Richardson scheme is consequently unconditionally unstable:

$$|G_{-}| > 1 \qquad \text{for } d > 0 \text{ and } \sin\left(\frac{\theta}{2}\right) \neq 0 \tag{68}$$

This result means that reducing the time step cannot eliminate the instability. Even when d is very small, one of the two amplification-factor roots remains greater than unity in magnitude for at least one nonzero Fourier mode.

Consequently, the Richardson scheme produces growing numerical disturbances and eventually leads to non-physical solutions. It is therefore unsuitable for practical transient diffusion simulations, including the present Plane Couette Flow problem.

F. Summary of the Richardson Scheme

The main properties of the explicit Richardson scheme are summarized as follows:

- It uses central differencing in time.
- It uses central differencing in space.
- It is second-order accurate in time.
- It is second-order accurate in space.
- It is an explicit three-time-level method.
- It is unconditionally unstable for the one-dimensional diffusion equation.
- It is not suitable for practical transient diffusion calculations.

Therefore, the Richardson scheme is included in this report as an important numerical example. It demonstrates that a scheme may possess a high formal order of accuracy while still being unsuitable because of its stability characteristics.